

Course Title: UCCD3113 Distributed Computer Systems

Assessment: Assignment (15%), Oral Presentation (5%)

Outcomes:

- Assignment (Report) - Construct distributed computing solution for specific applications (CLO4).
- Oral Presentation - Recommend the use of middleware, operating systems, integration methods and models for distributed systems solutions (CLO3).

1. Assignment Task

This assignment requires you to work collaboratively in a group of 4 members. Your group acts as a team of distributed system architects. You are required to propose a case study related to distributed computing, design, and deploy a simple system for your client's specific concerns (e.g., security, scalability, reliability, etc). **Note:** This is a design an architecture project.

The general flow of your task is as follows:

1. Propose a case study that relates to the selected domain. A more interesting and challenging problem to be solved is highly recommended.
2. Study and discuss in detail about the selected topic. Illustrate a working flow and diagram for the problem.
3. Refine your proposed solution with your tutor during tutorial sessions.
4. Produce a Technical Report for the overall problem and solution. Your report should include a background study and the intention of work on the selected topic.
5. In the report, include a manual (step-by-step with screenshots) in solving the proposed problem using the suggested Cloud platform by your course leader.
6. Refer to the Marking Rubric for the expectation on deliverables.
7. Each group need to provide an oral presentation for their work in Week 13-14.

Select a topic:

No	Topic
1	IoT
2	AI
3	Data Visualization
4	Database
5	Intelligent Service
6	Data Analytics
7	Storage

2. Group Formation & Management

Team Size: Strictly 4 members per group.

Leadership: Elect a Group Leader to handle submission and communication.

Group Registration: Groups must be formed latest by Week 2 using the link posted on Wble (see Week 1)

Deliverables: Each group is expected to deliver a project report (See Section 4.1, Deliverable A: Assignment Report) and oral presentation by group (See Section 4.2, Deliverable B: Oral Presentation).

3. Important Timeline

- Assignment Report Submission by Week 10 (Friday) - Upload to WBLE by Group Leader only.
- Oral Presentation on Week 13.

4. Deliverables

There are 2 deliverables. Your group is required to produce a comprehensive technical report and give oral presentation on your solution.

4.1 Deliverable A: Assignment Report (15%)

Format and Length: Submit your report in PDF format, 20 pages max (excl. references, cover page, and appendix). Your report must be structured as follows to align with the grading rubric. Download the report template on Wble.

Chapter 1: Introduction

- Case Study Definition: Clearly explain your selected case study. Describe the industry or domain related to it (e.g., Healthcare, E-Commerce, Fintech).
- Purpose: What is the primary goal of this system? What value does it bring?
- Real-World Context: Why does this system matter? Explain the real-world necessity (e.g., "Without this system, patient data is at risk of being lost in a fire", or "Slow loading times cause e-commerce sites to lose 40% of revenue").
- Target Users: Who will use this system? (e.g., "Doctors needing quick access to records", or "Shoppers looking for flash deals").

Chapter 2: Problem Definition

- Define the specific challenge your group is solving. Make this concise and clear.

Chapter 3: Methodology

- System Architecture Diagram: A professional, full-page DS Architecture Diagram showing how components connect.
- Data Flow Description: Walk the reader through your diagram step-by-step. Explain the purpose of the components included in the diagram.

Chapter 4: Evaluation and Discussion

- Benefits: Reflect on how your propose system deliver business value.
- What are the limitations of your specific design? Is it expensive? Is it complex to manage?
- Lessons Learned: Summary of key findings.
- Future Enhancements: Suggest possible improvements.

4.2. Deliverable B: Oral Presentation (5%)

Oral Presentation on Week 12. Please practice to ensure your presentation fits in 10 mins only. QnA 5 mins. (Oral Presentation + QnA = 15 mins). Since this is a design project, you need to walkthrough the Architecture. Slides: 8-10 Slides only.

Oral Presentation Outline:

- Problem Overview.
- The Architecture Walkthrough: Trace the path of a user request through your diagram. Explain how the components integrate to solve the specific problem stated.
- Show how you solved the specific challenge.
- Demonstrate the hosted website during the presentation session.
- Key Findings & Conclusion.

5. Assessment Rubric

Marks will be awarded according to the originality, comprehensiveness, and creativity of the discussion. The distribution of marks will follow the criteria below:

Assignment (Submit Technical Report as PDF)

- Successfully implements the proposed distributed/cloud-based solution for the chosen case study without supervision.
- Ability to implement the proposed solution completely and produce a comprehensive Technical Report and Full Manual with clear priorities in the design.
- Ability to show engagement and commitment in developing the case study solution and report.
- Ability to demonstrate several distributed/cloud components and their configuration for the chosen domain.
- Ability to organize a proper system/network architecture for the proposed case study problem.

Oral Presentation (By Group)

- Ability to present the case study, domain, and proposed distributed/cloud-based solution clearly and confidently.
- Perfect demonstration of system integration and live walkthrough of the implemented distributed/cloud solution (video presentation).

6. Frequently Asked Questions

Q: Can we form a group of 3 if we cannot find a fourth member?

A: No. All groups must have exactly 4 members. If you are short on members, you must check the shared group registration spreadsheet and contact other incomplete groups or unassigned individuals to merge. A group of 3 will only be authorized by the instructor at the end of the group formation deadline if the total number of students does not divide evenly by 4.

Q: Where and when do we register our group?

A: Groups must be registered no later than Week 2 using the provided Google Sheets link posted on Wble during Week 1.

Q: How do we choose our assignment topic?

A: You are given a broad domain (e.g., IoT, AI, Data Analytics). Choose 1 topic, and each group cannot have the same topic. You must post your chosen topic on the shared Google Sheet link so other groups can see it. Your group is then required to propose a specific, challenging case study and design an architecture within that assigned domain.

Q: What should we do if a group member is not contributing?

A: Professional collaboration is a core part of your assessment. You are expected to act as a professional team. Any internal disputes or non-contributing members must be managed and addressed diligently within the group.

Q: Do we need to build and code a fully working system, or is this just a design?

A: You must do both. This is an architecture design project, but you are also required to deploy a simple system to address your client's specific concerns. You must demonstrate the deployed system during your presentation and provide a step-by-step manual with screenshots in your report.

Q: What are the formatting rules for the technical report?

A: The report must be submitted in PDF format and cannot exceed 20 pages (excluding the cover page, references, and appendix). You must download and follow the structure provided in the report format link.

Q: How is the report submitted?

A: The Group Leader is responsible to upload the final PDF of the report to WBLE by Friday, Week 12.